



ANH TUAN CAO

Master of Civil Engineering

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INTRODUCTION AND RESEARCH OBJECTIVE

My name is Anh Tuan Cao. I graduated with a Master's degree in Construction Engineering in Korea in 2019, and I am currently working as a Project Engineer at Minh Lam Construction Company. After several years of professional experience to strengthen my technical expertise and practical knowledge, I am eager to continue my academic journey by pursuing a Ph.D. scholarship in Civil Engineering.

In particular, I am strongly interested in applying Artificial Intelligence, Machine Learning, Deep Learning to construction management. My research objective is to explore how AI-driven models can improve project planning, cost and schedule prediction, risk management, and overall decision-making. I aim to contribute to the advancement of intelligent construction management systems, enabling more efficient, sustainable, and data-driven practices in the built environment.

EDUCATION

► M.Sc. in Civil and Environmental Engineering - Kunsan National University, South Korea 2017 – 2019

- GPA: 4.31/4.5 (97.1%) Class rank: top 3
- Thesis: Optimal Vibration Control of Structure using Outrigger and Pendulum Tuned Mass Damper System.

► B.Sc. in Civil Engineering - Ho Chi Minh City University of Technology (HCMUT), Vietnam 2011 – 2016

- GPA: 8.66/10 (86.6%) Class rank: top 3
- Major: Civil and Industrial Structures

HONORS & AWARDS

- Best Presentation, The 6th Annual Conference of Vietnamese Young Scientists (ACVYS), 2019
- Certificate of Research Experience, 2019
- Outstanding Vietnamese Student Award, Kunsan National University, 2018
- Excellent Student Award, Ho Chi Minh City University of Technology, 2012 & 2015
- Encouragement Scholarship (every semester, 2011–2016)

RESEARCH AND WORKING EXPERIENCE

Construction Engineering Department, Minh Lam Construction Company 2024 – Present

Construction Engineering Department, Central Construction Joint Stock Company 2021 – 2024

Research Projects, South Korea 2017 – 2021

- Seismic risk assessment of Concrete Gravity Dam
- Seismic risk assessment of Nuclear Power Plant
- Vibration control of High-rise buildings subjected to seismic load
- Structural vibration analysis of Wind Turbines

Construction Engineering Department, NamCong Engineering Corporation 2016 – 2017

Construction Engineering Department, COFICO Joint Stock Company 2016 – 2016

RESEARCH INTERESTS

- Machine learning
- Deep learning
- Artificial intelligence

TECHNICAL & LANGUAGE SKILL

Technical Skills

- MS Office
- SAP2000, Etabs, Safe, Abaqus, Plaxis,
- AutoCAD, MS Project
- Matlab, Python

Language Skills

- Vietnamese: Native
- English: Fluent (IELTS 6.5)
- Korean: Very basic

PUBLICATIONS

Citations: 155

Link of Google Scholar: <https://scholar.google.com/citations?user=TODDZTkAAAAJ&hl=en&oi=srq>

TITLE

Cited by Year

Journal Papers (SCIE indexed):

- Cao, A.T. et al., Risk assessment of aged concrete gravity dam subjected to material deterioration under seismic excitation, **International Journal of Concrete Structures and Materials**. 26 2020
- Cao, A.T. et al., Seismic vulnerability of cabinet facility with tuned mass dampers subjected to high-and low-frequency earthquakes, **Applied Sciences**. 38 2020
- Cao, A.T. et al., Fragility Assessment for Electric Cabinet in Nuclear Power Plant Using Response Surface Methodology, **Nuclear Engineering and Technology**. 52 2019
- Cao, A.T. et al., Earthquake Risk Assessment of Concrete Gravity Dam by Cumulative Absolute Velocity and Response Surface Methodology, **Earthquakes and Structures**. 9 2019
- Cao, A.T. et al., Simplified Approach for Seismic Risk Assessment of Cabinet Facility in nuclear power plants based on cumulative absolute velocity, **Nuclear Technology**. 25 2019

Conference Proceedings:

- Influences of Ground Motion Frequency on Seismic Vulnerability of Nonstructural Components. 2021
- A Proposed Method for Inspecting and Predicting the Seismic Vulnerability of Dam Structures in Korea, ICSCEA. 2 2019
- Experimental and Numerical Modal Analysis of Cabinet Facility considering the connection nonlinearity, ICSCEA. 2 2019
- Optimization of Vibration Control using Outrigger and Pendulum Tuned Mass Damper System for High-Rise Building, RUET. 2019
- New Methodology for Predicting Seismic Risk of Concrete Gravity Dam, ACVYS. 2019
- VC4OWT: MATLAB Interface for Vibration Control of Offshore Wind Turbine, VSOE. 1 2018
- Strategy of Cumulative Absolute Velocity for Earthquake Risk Assessment of Concrete Gravity Dam